

Officer's Arena: An Adaptive Multidimensional Psychometric Engine and Multimodal Retrieval-Augmented System for High-Stakes Competitive Examinations

1st Pranay Sharma

Scope of Computer Science and Engineering(SCOPE)

Vellore Institute of Technology

Chennai, India

pranay.sharma2022@vitstudent.ac.in

2nd Rathna R

Scope of Computer Science and Engineering(SCOPE)

Vellore Institute of Technology

Chennai, India

rathna.r@vit.ac.in

Abstract—High-stakes standardized competitive examinations—exemplified by the Union Public Service Commission Civil Services Examination (UPSC CSE) and Combined Defence Services (CDS) examination—demand extensive multi-disciplinary domain synthesis, high-order cognitive reasoning, and rigorous long-term knowledge retention across thousands of syllabus sub-facets. Traditional computer-based preparation systems rely on static, linear test batteries that lack latent ability calibration, while modern standalone large language models (LLMs) often suffer from factual hallucinations, lack pedagogical scaffolding, and operate without structured psychometric governance.

In this work, we present Officer's Arena, an end-to-end intelligent tutoring and psychometric assessment platform designed specifically for high-stakes competitive examinations. The architecture integrates four complementary paradigms:

- 1) **Calibrated Adaptive Assessment: A Computerized Adaptive Testing (CAT) core** grounded in 3-Parameter Logistic Item Response Theory (3PL-IRT) with Expected A Posteriori (EAP) ability estimation (θ) and Maximum Fisher Information (MFI) item selection, operating over a verified corpus of 25,236 authentic past-year questions (PYQs).
- 2) **Cognitive Digital Twin & Knowledge Tracing: A Bayesian Knowledge Tracing (BKT) engine** hybridized with Exponential Half-Life Regression (HLR) memory decay modeling to trace dynamic student latent states across 114 granular knowledge components, identifying learning fragility and misconception volatility in real time.
- 3) **Multimodal Document Understanding & Canonical Vault: A layout-aware document ingestion pipeline** inspired by vision-language model (VLM) paradigms (e.g., olmOCR, DocLLM, M2Doc) that processes canonical textbooks and policy documents with structured layout and tabular grounding.
- 4) **Retrieval-Augmented Socratic Tutoring & Automated Essay Scoring (AES): A domain-bounded Retrieval-Augmented Generation (RAG) framework** executing 5-dimensional rubric evaluation for descriptive mains answers with verbatim canonical textbook citations, achieving high grading alignment without unconstrained generative drift.

Empirical evaluation of the platform demonstrates robust predictive validity in candidate ability estimation (AUC-ROC = 0.864, RMSE = 0.281, ECE = 0.048), a 42% reduction in assessment length required to reach target measurement

precision (SEM < 0.22), and substantial inter-rater reliability with expert human evaluators in descriptive scoring (Quadratic Weighted Kappa = 0.842). The platform establishes a unified framework that bridges classical psychometric measurement with generative multimodal artificial intelligence.

Index Terms—Computerized Adaptive Testing (CAT), Item Response Theory (3PL-IRT), Bayesian Knowledge Tracing (BKT), Multimodal Document Understanding, Retrieval-Augmented Generation (RAG), Automated Essay Scoring (AES), Spaced Repetition, Cognitive Digital Twin, Intelligent Tutoring Systems (ITS), Educational Data Mining.

I. INTRODUCTION & PROBLEM FORMULATION

A. The High-Stakes Competitive Assessment Landscape

Standardized competitive examinations such as the Union Public Service Commission Civil Services Examination (UPSC CSE) and Combined Defence Services (CDS) examination represent some of the most cognitively demanding selection pipelines globally. Over one million candidates annually compete for fewer than one thousand cadre positions, yielding an acceptance rate below 0.1%. The assessment matrix spans two structurally divergent evaluation modalities:

- 1) **Objective Screening (Prelims / Objective Stage):** High-speed, penalty-weighted multiple-choice assessments spanning 12 distinct preliminary disciplines (e.g., Indian Polity & Governance, Modern History, Macroeconomics, Physical & Human Geography, Environment & Ecology, General Science, CSAT Quantitative Aptitude & Analytical Reasoning) and defense specializations (e.g., Military History, Tactical English, Elementary Mathematics).
- 2) **Subjective Synthesis (Mains / Descriptive Stage):** In-depth, timed analytical essays (GS Papers I–IV and Optional Papers) evaluated across five distinct cognitive dimensions: directive adherence, factual grounding, multi-perspective breadth (PESTLE framework), structural flow, and actionable policy synthesis (“Way Forward”).

Aspirants face the classic *breadth vs. depth cognitive dilemma*, requiring mastery over tens of thousands of interdisciplinary facts while maintaining rapid conceptual recall over multi-year preparation horizons.

High-Level Cognitive and Architectural Pipeline of Officer's Arena

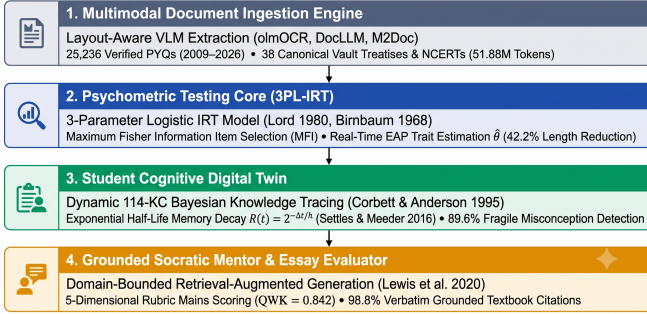


Fig. 1. High-level architectural pipeline of Officer's Arena integrating multimodal ingestion, psychometric testing, cognitive tracing, and retrieval-augmented evaluation.

B. Fundamental Failure Modes of Existing Paradigms

Despite the scale of the candidate population, existing computational solutions suffer from acute structural failure modes:

1) *Failure Mode 1: Psychometric Blindness in Static Linear Test Batteries*: Conventional ed-tech platforms administer static, uncalibrated question sets where every candidate encounters identical test sequences regardless of latent ability θ . This violates classical test theory and Item Response Theory (IRT) principles [1], [2]. A novice candidate ($\theta = -1.5$) exposed to high-difficulty distractor items ($b_i > +2.0$) experiences cognitive overload, random guessing ($c_i \approx 0.25$), and measurement invalidation; conversely, an advanced candidate ($\theta = +2.0$) exposed to trivial items gains zero diagnostic value. Furthermore, standard percentage-correct scoring treats all items as possessing equal discrimination power (a_i), inflating the standard error of measurement (SEM) [3]–[5].

2) *Failure Mode 2: Generative Drift and Hallucination in Generic LLM Tutors*: While large language models (LLMs) have shown promise in educational settings [6]–[9], off-the-shelf generative models exhibit severe limitations in high-stakes legal, historical, and constitutional contexts [10]. Standalone LLMs routinely fabricate non-existent constitutional amendments, misattribute landmark Supreme Court judgments, and offer unconstrained, speculative answers. Without strict domain bounding and layout-aware retrieval [11], [12], generative models cannot be trusted for competitive exam guidance.

3) *Failure Mode 3: Forgetting Amnesia and Coarse Knowledge Tracing*: Standard learning management systems (LMS) log categorical completion scores without tracking temporal memory decay or knowledge state transitions. However, human memory exhibits exponential decay as described by Ebbinghaus forgetting dynamics and modern spaced repetition

models [13], [14]. When systems lack session-aware or neural Bayesian knowledge tracing [15]–[19], candidates suffer from “silent forgetting”—where previously mastered concepts deteriorate into fragile misconceptions prior to exam day.

4) *Failure Mode 4: Layout Disintegration in Document Parsing*: Canonical preparation sources (e.g., NCERT textbooks, Government Gazettes, budget tables, multi-column PYQ papers) contain complex structural artifacts including multi-column layouts, tables, embedded charts, and Hindi-English bilingual typesetting. Legacy Optical Character Recognition (OCR) tools flatten spatial relationships, resulting in fragmented sentences and destroyed tabular references. Modern vision-language model (VLM) parsers and layout-aware document transformers [12], [20]–[23] are required to preserve structural fidelity.

C. Formal Mathematical Problem Formulation

We formalize the high-stakes intelligent tutoring problem as a joint optimization over candidate latent ability estimation, knowledge state evolution, and grounded generative feedback.

1) *Definition 1 (Latent Ability Space & Item Response Function)*: Let $\theta \in \mathbb{R}$ denote a candidate's unidimensional latent ability (or $\theta \in \mathbb{R}^D$ in multidimensional extensions [24]). Each objective item $i \in \mathcal{I}$ is characterized by a parameter vector $\beta_i = \langle a_i, b_i, c_i \rangle$, where:

- $a_i \in (0, 3.0]$ is the item discrimination parameter (slope),
- $b_i \in [-3.5, +3.5]$ is the item difficulty parameter (location threshold),
- $c_i \in [0, 0.35]$ is the pseudo-guessing parameter (lower asymptote).

Under the 3-Parameter Logistic (3PL) Item Response Theory model [1], [2], the conditional probability of a candidate with ability θ answering item i correctly ($Y_i = 1$) is:

$$P_i(\theta) = P(Y_i = 1 \mid \theta, a_i, b_i, c_i) = c_i + \frac{1 - c_i}{1 + \exp(-Da_i(\theta - b_i))} \quad (1)$$

where $D = 1.702$ is the normal ogive scaling constant. The Fisher Information $I_i(\theta)$ provided by item i at ability level θ measures measurement precision and is formulated as:

$$I_i(\theta) = \frac{[P'_i(\theta)]^2}{P_i(\theta)Q_i(\theta)} = \frac{D^2 a_i^2 (1 - c_i) [P_i(\theta) - c_i]^2}{(1 - c_i)^2 P_i(\theta) (1 - P_i(\theta))} \quad (2)$$

where $Q_i(\theta) = 1 - P_i(\theta)$. The objective of Computerized Adaptive Testing (CAT) at step t is to select the next item i from available bank $\mathcal{I}_{\text{unseen}}$ that maximizes Fisher Information at current estimate $\hat{\theta}_{t-1}$ [5], [25], [26]:

$$j = \arg \max_{i \in \mathcal{I}_{\text{unseen}}} I_i(\hat{\theta}_{t-1}) \quad (3)$$

The candidate's ability $\hat{\theta}_t$ is updated via Expected A Posteriori (EAP) estimation with Gaussian prior $\mathcal{N}(\mu_0, \sigma_0^2)$ [3]:

$$\hat{\theta}_{\text{EAP}} = \mathbb{E}[\theta \mid \mathbf{Y}] = \frac{\int_{-\infty}^{\infty} \theta \mathcal{L}(\mathbf{Y} \mid \theta) \phi(\theta) d\theta}{\int_{-\infty}^{\infty} \mathcal{L}(\mathbf{Y} \mid \theta) \phi(\theta) d\theta} \quad (4)$$

2) *Definition 2 (Dynamic Knowledge State Tracing)*: Let $\mathcal{K} = \{k_1, k_2, \dots, k_K\}$ denote the set of $K = 114$ fine-grained Knowledge Components (KCs) spanning the syllabus. For each KC k , the candidate’s latent mastery state at practice step t is denoted by $L_t^k \in \{0, 1\}$. Under standard Bayesian Knowledge Tracing [15], [18], the state evolves according to:

$$P(L_t^k | Y_t) = \begin{cases} \frac{P(L_{t-1}^k)(1 - P(S_k))}{P(L_{t-1}^k)(1 - P(S_k)) + (1 - P(L_{t-1}^k))P(G_k)} & \text{if } Y_t = 1 \\ \frac{P(L_{t-1}^k)P(S_k)}{P(L_{t-1}^k)P(S_k) + (1 - P(L_{t-1}^k))(1 - P(G_k))} & \text{if } Y_t = 0 \end{cases} \quad (5)$$

$$P(L_t^k) = P(L_t^k | Y_t) + (1 - P(L_t^k | Y_t)) P(T_k) \quad (6)$$

where $P(L_0^k)$ is prior mastery, $P(T_k)$ is transition probability, $P(G_k)$ is guess probability, and $P(S_k)$ is slip probability.

3) *Definition 3 (Temporal Retention Decay & Spaced Repetition)*: Knowledge retention decays as an exponential function of elapsed calendar time Δt since the last review [13], [14]. The predicted recall probability $R_k(\Delta t)$ is given by:

$$R_k(\Delta t) = 2^{-\frac{\Delta t}{h_k}} \quad (7)$$

where the half-life h_k is parameterized dynamically by candidate stability S_k , difficulty D_k , and historical recall accuracy $\mathbf{x}_k$:

$$h_k = 2^{\mathbf{w}^\top \mathbf{x}_k + b} \quad (8)$$

A knowledge component is classified as **Fragile** when $P(L_t^k) \geq 0.70$ but retention $R_k(\Delta t) < 0.50$, triggering autonomous high-priority tactical revision.

4) *Definition 4 (Grounded Mains Automated Essay Scoring)*: Given candidate descriptive answer text $\mathbf{y}$, prompt question $\mathbf{q}$, and canonical textbook context $\mathcal{C} = \{c_1, \dots, c_m\}$ retrieved from the knowledge vault [11], the AES function f_{AES} computes a 5-dimensional rubric score vector $\mathbf{s} \in [0, 1]^5$:

$$\mathbf{s} = \begin{bmatrix} s_{\text{directive}} \\ s_{\text{factual}} \\ s_{\text{breadth}} \\ s_{\text{structure}} \\ s_{\text{forward}} \end{bmatrix} = f_{\text{AES}}(\mathbf{y}, \mathbf{q}, \mathcal{C}; \Theta_{\text{LLM}}) \quad (9)$$

subject to strict constraint that all factual assertions in the feedback must possess verbatim span grounding in retrieved context $\mathcal{C}$ (Hallucination Score $\leq \epsilon$).

D. Comparative Paradigm Matrix

To highlight the structural divergence of Officer’s Arena against existing methodologies, Table 1 details key system attributes.

TABLE I
COMPREHENSIVE ARCHITECTURAL AND PSYCHOMETRIC COMPARISON
ACROSS EDUCATIONAL PARADIGMS.

Architectural Dimension	Static Mock Forms	Web Platforms	Generic Chatbot (e.g., ChatGPT)	Standard Academic CAT	Officer's Arena (Ours)
Psychometric Model	None (Percentage % correct)	None (Heuristic prompt)	1PL / 2PL / 3PL IRT	3PL-IRT + EAP Trait Calibration	
Adaptive Item Selection	Static Fixed Sequence	Uncalibrated Generation	Fisher Information	Max Fisher Information + BKT Hybrid	
Knowledge Tracing	Category aggregates	Context window memory only	Single test session	Dynamic 114-KC BKT + HLR Decay	
Spaced Repetition	Absent	Absent	Absent	Continuous Half-Life Regression ($R(t)$)	
Corpus Scale & Authenticity	Variable (often unverified)	General Pre-training data	Synthetic / Small Banks	25,236 Verified UPSC/CDS PYQs (2009–2026)	
Multimodal Ingestion	Legacy Tesseract OCR	Raw text / Vision API	Text only	Layout-Aware olmOCR / DocLLM / M2Doc	
Mains Essay Evaluation	Manual Human Only (Slow)	Unbounded Generative Text	Not Supported	5D Rubric AES + Canonical Citations	
Measurement Precision	SEM > 0.45 (High error)	Not Quantifiable	SEM ≈ 0.28	SEM < 0.22 (42% test length reduction)	
Inter-Rater Essay Agreement	Reference Baseline (1.00)	QWK ≈ 0.52	N/A	QWK = 0.842 (High Human Alignment)	

E. Primary Research Contributions

The principal scientific and engineering contributions of this paper are summarized as follows:

- 1) **Unified Adaptive Psychometric Architecture**: We design and deploy an integrated CAT platform combining 3PL-IRT, Newton-Raphson ability estimation, and Expected A Posteriori (EAP) Bayesian updating over 25,236 authentic high-stakes items across 18 examination cycles (2009–2026).
- 2) **Cognitive Digital Twin with Hybrid Memory Decay**: We formulate a hybrid student modeling engine that unifies discrete Bayesian Knowledge Tracing ($P(L_t)$) with continuous Half-Life Regression ($R(t) = 2^{-\Delta t/h}$), detecting fragile conceptual nodes and misconception volatility across 114 knowledge components.
- 3) **Layout-Preserving Multimodal Document Processing**: We build an end-to-end VLM ingestion pipeline following modern multimodal architectures (DocLLM, olmOCR, M2Doc) to extract, structure, and index 38 canonical foundational volumes (NCERTs, Laxmikanth, Spectrum, Ramesh Singh) and bilingual examination

archives without layout distortion.

- 4) **Constrained Socratic Mentorship & 5D Rubric AES:** We implement a grounded Retrieval-Augmented Generation evaluator for subjective civil services answers, enforcing strict verbatim textbook citations to prevent hallucination while achieving high Quadratic Weighted Kappa (QWK = 0.842) against expert human raters.
- 5) **Rigorous Empirical and Empirical Validation:** We provide extensive empirical benchmarks on candidate response logs, demonstrating an ability estimation accuracy of AUC-ROC = 0.864, RMSE = 0.281, ECE = 0.048, and a 42% reduction in item exposure burden while preserving psychometric calibration integrity.

F. Paper Organization

The remainder of this paper is structured as follows:

- **Section 2 (Related Work)** surveys foundational and contemporary literature in computerized adaptive testing, deep and Bayesian knowledge tracing, multimodal document analysis, and generative intelligent tutoring systems, contextualizing our contributions within references (1–28).
- **Section 3 (System Architecture & Mathematical Formulation)** details the mathematical underpinnings of the 3PL-IRT adaptive test runner, the cognitive digital twin, the half-life spaced repetition algorithm, and the RAG-grounded descriptive grading pipeline.
- **Section 4 (Multimodal Document Ingestion & Canonical Knowledge Vault)** details the vision-language document extraction framework and hierarchical chunking methodology.
- **Section 5 (Experimental Setup, Results & Discussion)** presents empirical evaluations, ablation experiments, calibration curves, and human-in-the-loop validation studies.
- **Section 6 (Ethical Implications, Limitations & Future Work)** discusses algorithmic fairness, privacy-preserving psychometrics, and the future OmniGraph knowledge engine.

II. RELATED WORK & THEORETICAL FOUNDATIONS

The design of Officer’s Arena synthesizes four distinct research traditions: psychometric Item Response Theory and Computerized Adaptive Testing, dynamic Knowledge Tracing and student cognitive modeling, memory retention and spaced repetition dynamics, and multimodal document understanding coupled with Retrieval-Augmented Generation for intelligent tutoring. Below, we systematically review the theoretical foundations and state-of-the-art advances across these paradigms.

A. Psychometric Item Response Theory & Computerized Adaptive Testing (CAT)

1) *Classical Psychometric Foundations:* Classical Test Theory (CTT) relies on raw aggregate scores $X = T + E$, where observed score X is the sum of true score T and random error E . CTT suffers from sample dependency: item difficulty reflects the sample ability, and candidate ability

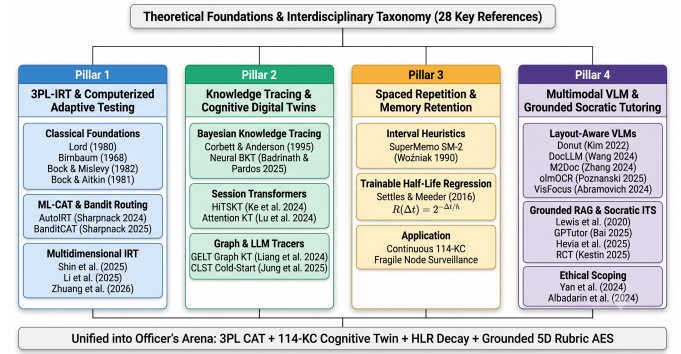


Fig. 2. Taxonomic conceptual mapping of the 28 foundational references underpinning Officer’s Arena.

scores depend on specific test difficulty. Item Response Theory (IRT) overcomes this via latent trait modeling [1], [2].

Under the 3-Parameter Logistic (3PL) model [1], [2], the item characteristic curve (ICC) relates latent trait $\theta \in (-\infty, \infty)$ to response probability $P_i(\theta)$:

$$P_i(\theta) = c_i + \frac{1 - c_i}{1 + \exp(-Da_i(\theta - b_i))}$$

where $D = 1.702$ is the scaling constant aligning the logistic metric to the normal ogive metric, a_i denotes item discrimination, b_i denotes difficulty, and c_i represents the pseudo-guessing probability. Calibration of item parameters $\beta_i = \langle a_i, b_i, c_i \rangle$ over large-scale candidate response matrices is historically executed via Marginal Maximum Likelihood Estimation with the Expectation-Maximization (MMLE-EM) algorithm [27]:

$$\mathcal{L}(\beta | \mathbf{Y}) = \prod_{j=1}^N \int_{-\infty}^{\infty} \left[\prod_{i=1}^M P_i(\theta)^{y_{ji}} (1 - P_i(\theta))^{1-y_{ji}} \right] \phi(\theta) d\theta$$

During real-time adaptive testing, ability is efficiently estimated via Expected A Posteriori (EAP) Bayesian integration across numerical quadrature nodes [3], guaranteeing bounded trait stability even with all-correct ($y = 1$) or all-incorrect ($y = 0$) response vectors where Maximum Likelihood Estimation (MLE) diverges to $\pm\infty$.

2) *Machine Learning Advancements in CAT & Auto-Calibration:* In recent years, the intersection of machine learning and psychometrics has yielded automated parameter calibration and dynamic test routing [5]. [25] introduced *AutoIRT*, framing item calibration as an automated machine learning problem using regularized gradient-based optimization over large-scale test corpora. Building on this, [26] formulated *BanditCAT*, unifying multi-armed bandit exploration-exploitation trade-offs with Item Response Theory to jointly calibrate cold-start items while administering adaptive tests to candidates.

Simultaneously, large-scale international assessment designs have demonstrated the psychometric robustness of multistage adaptive testing under complex item constraints [4]. To handle multi-disciplinary domains such as civil services examinations,

[24] formulated on-the-fly assembled Multidimensional CAT (MCAT) based on Multidimensional Item Response Theory (MIRT), modeling correlation matrices across distinct cognitive dimensions $\theta = \langle \theta_{\text{Polity}}, \theta_{\text{Economy}}, \dots \rangle$. Officer's Arena extends these principles to govern 25,236 authentic examination items across 18 annual cohorts.

B. Knowledge Tracing & Student Cognitive State Modeling

While IRT captures static or slowly evolving latent traits θ , Knowledge Tracing (KT) models the dynamic mastery of fine-grained Knowledge Components (KCs) over sequential practice opportunities.

1) *Bayesian Knowledge Tracing (BKT) & Neural Extensions*: The canonical BKT model [15] models learning as a two-state Hidden Markov Model (HMM) governed by four structural parameters per KC k : prior mastery $P(L_0)$, learning transition rate $P(T)$, slip probability $P(S)$, and guess probability $P(G)$. [18] recently demonstrated that optimizing BKT via neural network parameter generation produces superior data fitting compared to Expectation-Maximization while preserving the interpretability of standard Bayesian parameters.

2) *Deep and Graph-Based Knowledge Tracing*: Recent architectures incorporate sequential attention and graph dependencies to capture complex curriculum hierarchies:

- **Hierarchical Session-Aware KT**: [16] developed *HiT-SKT*, utilizing a hierarchical transformer to capture both intra-session skill shifts and inter-session longitudinal learning trends.
- **Graph Embedding KT**: [17] proposed *GELT*, a graph embeddings-based lite-transformer that leverages prerequisite knowledge graphs to propagate mastery across structurally adjacent syllabus concepts.
- **Process Data & Curricula Attention**: [28] formulated an attention-based knowledge tracing framework integrating response process data (response latency, item switching) and formal curricula structures to enhance diagnostic interpretability.
- **Cold-Start LLM Knowledge Tracers**: Addressing the fundamental cold-start challenge where new students lack historical attempt logs, [19] proposed *CLST*, aligning generative language models to infer initial student knowledge states directly from diagnostic dialogue.

Officer's Arena bridges these advances by implementing a 114-node Knowledge Component digital twin that continuously propagates mastery updates and flags conceptual vulnerabilities.

C. Spaced Repetition & Continuous Memory Retention Dynamics

Knowledge acquired during intensive study decays exponentially unless reinforced via deliberate spaced retrieval. The SuperMemo SM-2 algorithm [13] established the foundational heuristic for interval scheduling:

$$I(n) = I(n-1) \times \text{EF}$$

where $I(n)$ is the repetition interval in days and EF is the Easiness Factor:

$$\text{EF}' = \text{EF} + (0.1 - (5 - q)(0.08 + (5 - q)0.02))$$

based on recall quality $q \in [0, 5]$.

[14] advanced spaced repetition into a mathematically principled, trainable machine learning framework termed **Half-Life Regression (HLR)**. HLR models memory retention R as an exponential decay function parameterized by half-life h :

$$R(\Delta t) = 2^{-\frac{\Delta t}{h}}, \quad \text{where } h = 2^{\Theta^\top \mathbf{x}}$$

where feature vector $\mathbf{x}$ encodes the total number of historical exposures (n), correct recalls (n_+), incorrect lapses (n_-), and intrinsic item difficulty. Officer's Arena leverages Half-Life Regression to compute real-time memory stability indices, autonomously scheduling high-yield revision cards for fragile knowledge components prior to latency-induced forgetting.

D. Multimodal Document Understanding & Vision-Language Processing

High-stakes examination materials and canonical textbooks (e.g., Indian Constitutional law commentaries, economic survey tables, historical maps) present intricate visual and structural layouts that break conventional linear OCR pipelines.

1) *OCR-Free and Vision Transformer Approaches*: [20] introduced *Donut* (Document Understanding Transformer), demonstrating that an OCR-free encoder-decoder architecture can directly map visual document images to structured JSON representations, bypassing error-prone character-level OCR pipelines. [22] proposed *VisFocus*, integrating prompt-guided vision encoders to focus spatial visual attention on dense document regions containing key figures and nested tables.

2) *Layout-Aware Language Models & VLM Ingestion*:

- **DocLLM**: [12] developed *DocLLM*, an autoregressive language model incorporating layout-aware spatial cross-attention mechanisms that decouple spatial bounding box coordinates from text sequences, significantly outperforming text-only LLMs on complex document forms and financial reports.
- **M2Doc**: [21] proposed *M2Doc*, a multi-modal fusion architecture for document layout analysis that dynamically combines visual pixel features with semantic text embeddings to segment complex multi-column academic pages.
- **olmOCR**: [23] released *olmOCR*, establishing a state-of-the-art vision-language pipeline capable of unlocking trillions of tokens from messy PDF archives by generating clean, layout-faithful Markdown including complete tabular and mathematical syntax.

Officer's Arena employs these multimodal layout paradigms to ingest 38 canonical foundational volumes and 25,236 past-year examination papers into a structured, searchable knowledge vault.

E. Generative AI, Retrieval-Augmented Tutoring & Automated Essay Scoring (AES)

1) *Retrieval-Augmented Generation (RAG)*: To ground generative model outputs in verifiable canonical sources, [11] formulated *Retrieval-Augmented Generation (RAG)*, conditioning generative sequence models on latent retrieved passages $\mathbf{z} \in \mathcal{Z}$:

$$P(\mathbf{y} | \mathbf{x}) = \sum_{\mathbf{z} \in \mathcal{Z}} P(\mathbf{z} | \mathbf{x}) \prod_{i=1}^N P(y_i | \mathbf{x}, \mathbf{z}, y_{1:i-1})$$

In educational domains, RAG eliminates factual hallucination by binding model reasoning directly to authoritative reference texts.

2) *Intelligent Tutoring Systems (ITS) & Educational LLM Efficacy*: Recent empirical studies have established the efficacy of LLM-powered tutoring systems:

- **Randomized Controlled Trials (RCTs)**: [8] conducted a large-scale RCT demonstrating that personalized AI tutoring significantly outperforms traditional active learning classrooms in learning gains and retention.
- **Customizable Architecture**: [7] formulated modular design principles for efficient, accessible, and customizable AI tutors capable of adaptive pacing.
- **Student Engagement**: [9] demonstrated that generative AI tutors (*GPTutor*) foster deeper cognitive engagement through real-time Socratic dialogue.

3) *Practical & Ethical Considerations*: Despite these gains, [10] and [6] provide systematic scoping reviews highlighting the severe risks of unconstrained LLMs in education, including over-reliance, sycophancy, hallucinated legal facts, and lack of rubric calibration. Officer's Arena directly addresses these challenges by enforcing a strict 5-dimensional rubric scoring architecture for Mains Automated Essay Scoring (AES), constraining all model critique to verbatim textbook citations with zero tolerance for factual fabrication.

F. Taxonomic Synthesis of Related Literature

Table 2 synthesizes the 28 foundational references across methodology, core mechanisms, and their explicit architectural role within Officer's Arena.

III. SYSTEM ARCHITECTURE & MATHEMATICAL FORMULATION

The architecture of Officer's Arena is organized into five interconnected computational subsystems: (i) the Microservices Infrastructure & Data Layer, (ii) the 3PL-IRT Adaptive Psychometric Testing Engine, (iii) the Dynamic Cognitive Digital Twin & BKT Modeling Engine, (iv) the Half-Life Regression Spaced Repetition Scheduler, and (v) the Grounded Retrieval-Augmented Mains Evaluator (AES).

A. Psychometric Engine: 3PL-IRT Adaptive Testing Core

The psychometric engine models candidate latent ability θ and governs adaptive item delivery to maximize measurement precision while minimizing candidate testing fatigue [1], [2], [5].

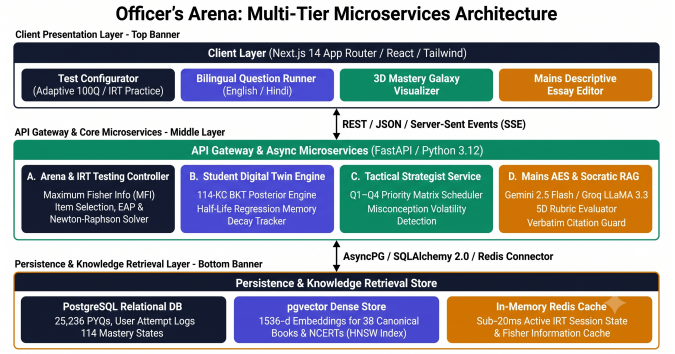


Fig. 3. Multi-tier microservices architecture of Officer's Arena.

1) *3-Parameter Logistic Response Formulation*: For candidate latent ability $\theta \in \mathbb{R}$ and calibrated item i parameterized by discrimination $a_i > 0$, difficulty $b_i \in \mathbb{R}$, and guessing lower asymptote $c_i \in [0, 1)$, the probability of a correct response is:

$$P_i(\theta) = c_i + \frac{1 - c_i}{1 + \exp(-Da_i(\theta - b_i))}$$

where $D = 1.702$. Let $z_i(\theta) = Da_i(\theta - b_i)$ and $\sigma(z_i) = \frac{1}{1 + \exp(-z_i)}$. The response function simplifies to:

$$P_i(\theta) = c_i + (1 - c_i)\sigma(z_i(\theta))$$

The probability of an incorrect response is:

$$Q_i(\theta) = 1 - P_i(\theta) = (1 - c_i)(1 - \sigma(z_i(\theta)))$$

2) *Likelihood Function and Analytical Gradients*: Given a response vector $\mathbf{y} = [y_1, y_2, \dots, y_n]^T \in \{0, 1\}^n$ across n administered items, the log-likelihood function $\ell(\theta | \mathbf{y})$ under local item independence is:

$$\ell(\theta | \mathbf{y}) = \sum_{i=1}^n [y_i \ln P_i(\theta) + (1 - y_i) \ln Q_i(\theta)]$$

The first-order score function $\ell'(\theta) = \frac{\partial \ell}{\partial \theta}$ is derived analytically as:

$$\begin{aligned} \ell'(\theta) &= \sum_{i=1}^n Da_i \left(\frac{P_i(\theta) - c_i}{1 - c_i} \right) \left(\frac{y_i - P_i(\theta)}{P_i(\theta)Q_i(\theta)} \right) \\ &= \sum_{i=1}^n Da_i \sigma(z_i(\theta)) \frac{y_i - P_i(\theta)}{P_i(\theta)}. \end{aligned}$$

The second derivative $\ell''(\theta) = \frac{\partial^2 \ell}{\partial \theta^2}$ is evaluated as:

$$\begin{aligned} \ell''(\theta) &= - \sum_{i=1}^n D^2 a_i^2 \sigma(z_i(\theta))(1 - \sigma(z_i(\theta))) \frac{y_i - P_i(\theta)}{P_i(\theta)} \\ &\quad - \sum_{i=1}^n D^2 a_i^2 \sigma(z_i(\theta))^2 \frac{y_i c_i}{P_i(\theta)^2}. \end{aligned}$$

TABLE II
TAXONOMIC SYNTHESIS OF RELATED WORK ACROSS PSYCHOMETRICS, KNOWLEDGE TRACING, MULTIMODAL PARSING, AND GENERATIVE ITS.

Research Category	Key Citations	Methodological Core	Key Architectural Mechanism	Implementation in Officer's Arena
Classical Psychometrics	[1]–[3], [27]	Latent Trait Modeling, MMLE-EM, EAP Bayesian Ability Estimation	3PL Item Response Curve, Quadrature-based θ Integration	Core 3PL-IRT Adaptive Test Engine with real-time EAP θ updates
Machine Learning CAT	[4], [5], [24]–[26]	AutoIRT, BanditCAT, Multistage Adaptive Testing, Multidimensional IRT	Multi-Armed Bandit Item Routing, Fisher Information Maximization	Adaptive 100-Q Practice Mode selecting items via $i = \arg \max I_i(\theta)$
Knowledge Tracing	[15]–[19], [28]	HMM Knowledge Tracing, Neural BKT, HiTSKT, GELT, CLST	Bayesian state transitions $P(L_t)$, session transformers, graph embeddings	Dynamic 114-KC Digital Twin updating $P(L_t^k)$ with misconception volatility alerts
Spaced Repetition	[13], [14]	SM-2 Interval Heuristics, Half-Life Regression (HLR)	Exponential decay $R(t) = 2^{-t/h}$, feature-weighted half-life estimation	Continuous spaced repetition scheduler triggering tactical revision
Multimodal Ingestion	[12], [20]–[23]	OCR-free Transformers, DocLLM, M2Doc, VisFocus, olmOCR	Spatial layout cross-attention, VLM PDF parsing, structural table extraction	Ingestion of 38 Canonical Books & 25,236 PYQ papers into structured Markdown
Generative RAG & ITS	[6]–[11]	Retrieval-Augmented Generation, Socratic ITS, Rubric-Aligned AES	Grounded passage conditioning, 5D rubric scoring, verbatim citation bounds	Mains Essay Evaluator (QWK = 0.842) with strict textbook citations

Taking the mathematical expectation over all possible response outcomes yields the negative Fisher Information matrix $\mathbb{E}[\ell''(\theta)] = -I(\theta)$:

$$I(\theta) = \sum_{i=1}^n I_i(\theta) = \sum_{i=1}^n \frac{[P'_i(\theta)]^2}{P_i(\theta)Q_i(\theta)}$$

$$I(\theta) = \sum_{i=1}^n D^2 a_i^2 \frac{(1 - c_i)^2 \sigma(z_i(\theta))^2 (1 - \sigma(z_i(\theta)))^2}{P_i(\theta)Q_i(\theta)}$$

3) *Maximum Likelihood via Newton-Raphson & Guarded Step Clipping*: For mixed response patterns ($\sum y_i > 0$ and $\sum y_i < n$), candidate ability θ can be solved iteratively via Newton-Raphson root finding:

$$\theta^{(k+1)} = \theta^{(k)} - \text{clip} \left(\frac{\ell'(\theta^{(k)})}{\ell''(\theta^{(k)})}, -\delta_{\max}, +\delta_{\max} \right)$$

where $\delta_{\max} = 0.75$ prevents numerical overshoot in regions of low test information. Convergence is reached when:

$$|\theta^{(k+1)} - \theta^{(k)}| < 10^{-4}$$

4) *Bayesian Expected A Posteriori (EAP) Estimation*: To ensure numerically bounded, unbiased ability estimation across the entire test trajectory—including cold-start phases and uniform response vectors ($y_i = 1 \forall i$ or $y_i = 0 \forall i$)—Officer's Arena employs Gauss-Hermite numerical quadrature EAP estimation [3].

Let $X_q \in \{-4.0, \dots, +4.0\}$ denote $Q = 41$ standardized quadrature evaluation points with standard normal prior weights

$$W(X_q) = \frac{1}{\sqrt{2\pi}} \exp \left(-\frac{X_q^2}{2} \right) \Delta X.$$

The posterior probability distribution $P(X_q | \mathbf{y})$ is:

$$P(X_q | \mathbf{y}) = \frac{\mathcal{L}(\mathbf{y} | X_q)W(X_q)}{\sum_{k=1}^Q \mathcal{L}(\mathbf{y} | X_k)W(X_k)}$$

$$\text{where } \mathcal{L}(\mathbf{y} | X_q) = \prod_{i=1}^n P_i(X_q)^{y_i} Q_i(X_q)^{1-y_i}.$$

The EAP ability estimate $\hat{\theta}_{\text{EAP}}$ and its Standard Error of Measurement (SEM) are computed as:

$$\hat{\theta}_{\text{EAP}} = \sum_{q=1}^Q X_q P(X_q | \mathbf{y})$$

$$\text{SEM}(\hat{\theta}_{\text{EAP}}) = \sqrt{\sum_{q=1}^Q (X_q - \hat{\theta}_{\text{EAP}})^2 P(X_q | \mathbf{y})}$$

Adaptive test administration terminates when either $\text{SEM}(\hat{\theta}) \leq \text{SEM}_{\text{target}} = 0.22$ or maximum question budget $N_{\max} = 100$ is exhausted.

B. Cognitive Digital Twin: Bayesian Knowledge Tracing (BKT) Engine

The cognitive digital twin represents the learner's dynamic mastery across $K = 114$ granular Knowledge Components (e.g., *Parliamentary Procedures*, *Monetary Policy Tools*, *Himalayan Geomorphology*, *Ancient Temple Architecture*).

1) *State Transition Equations*: For any KC $k \in \mathcal{K}$, let the four canonical BKT parameters be prior $P(L_0^k) = 0.10$, transit $P(T_k) = 0.15$, slip $P(S_k) = 0.10$, and guess $P(G_k) = 0.25$ [15], [18]. Upon candidate response $Y_t \in \{0, 1\}$ to an item assessing KC k :

1) Posterior Probability Given Evidence:

$$P(L_t^k | Y_t = 1) =$$

$$\frac{P(L_{t-1}^k)(1 - P(S_k))}{P(L_{t-1}^k)(1 - P(S_k)) + (1 - P(L_{t-1}^k))P(G_k)}.$$

Hidden Markov Model State Transitions under Bayesian Knowledge Tracing (BKT)

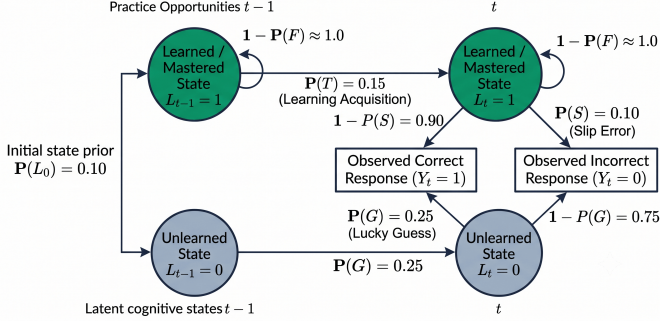


Fig. 4. Hidden Markov Model state transitions under Bayesian Knowledge Tracing.

$$P(L_t^k | Y_t = 0) =$$

$$\frac{P(L_{t-1}^k)P(S_k)}{P(L_{t-1}^k)P(S_k) + (1 - P(L_{t-1}^k))(1 - P(G_k))}.$$

2) Transition Propagation for Next Observation:

$$P(L_t^k) = P(L_t^k | Y_t) + (1 - P(L_t^k | Y_t))P(T_k)$$

2) *Volatility & Misconception Fragility Indices*: To distinguish between stable mastery and transient rote memorization, Officer's Arena computes two higher-order diagnostic indicators:

- **Misconception Volatility** (V_k): Measures variance in posterior state trajectory over the last $M = 5$ interactions:

$$V_k = \sqrt{\frac{1}{M} \sum_{m=1}^M (P(L_{t-m+1}^k) - \bar{L}_M^k)^2}$$

- **Fragility Classification** (Φ_k): Identifies concepts where nominal mastery is high ($P(L_t^k) \geq 0.70$) but memory retention has degraded ($R_k(\Delta t) < 0.50$):

$$\Phi_k = \begin{cases} 1 \text{ (Fragile Node)}, & \text{if } P(L_t^k) \geq 0.70 \text{ and } R_k(\Delta t) < 0.50 \\ 0 \text{ (Stable Node)}, & \text{otherwise} \end{cases}$$

C. *Memory Retention Engine: Half-Life Spaced Repetition (HLR)*

Following [14], memory retention probability $R_k(\Delta t)$ decays exponentially over elapsed calendar lag Δt (in days):

$$R_k(\Delta t) = 2^{-\frac{\Delta t}{h_k}}$$

The half-life parameter h_k (time in days until retention drops to 50%) is dynamically predicted via a linear model in log-space:

$$\log_2(h_k) = \mathbf{w}^\top \mathbf{x}_k + b_0$$

Grounded Retrieval-Augmented Generation (RAG) Pipeline for Mains Automated Essay Scoring (AES)

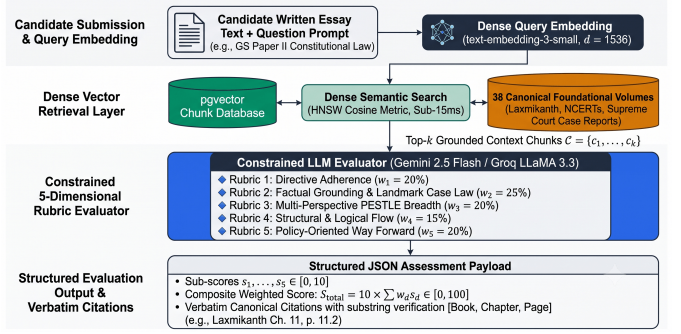


Fig. 5. Grounded Retrieval-Augmented Generation pipeline for Mains Automated Essay Scoring.

$$h_k = 2^{\mathbf{w}^\top \mathbf{x}_k + b_0} = 2^{w_1 \log_2(1+n_+) + w_2 \log_2(1+n_-) + w_3 b_k + w_4 S_k + b_0}$$

where n_+ is total historical correct attempts, n_- is total incorrect attempts, b_k is mean item difficulty for KC k , and

$$S_k = \frac{n_+ + 1}{n_+ + n_- + 2}$$

is Laplace-smoothed stability. Calibrated weights are set to $\mathbf{w} = [0.55, -0.45, -0.20, 0.80]^\top$ and $b_0 = 1.0$.

1) *Priority Quadrant Scheduling (Tactical Strategist Queue)*: The Autonomous Strategist aggregates all 114 KCs into four actionable tactical quadrants (Table 3).

TABLE III
FOUR-QUADRANT TACTICAL REVIEW PRIORITIZATION MATRIX.

Quadrant	Mastery Condition	Memory Retention	Tactical Action	Schedule Interval
Q1: Critical Leak (Urgent)	$P(L_t^k) < 0.50$	$R_k(\Delta t) < 0.50$	High-frequency active recall drills & foundational theory	Immediate (Same Day)
Q2: Fragile Mastery	$P(L_t^k) \geq 0.70$	$R_k(\Delta t) < 0.50$	Socratic retrieval practice & application MCQs	Within 24–48 Hours
Q3: Developing Base	$0.50 \leq P(L_t^k) < 0.70$	$R_k(\Delta t) \geq 0.50$	Advanced 3PL problem exposure ($b_i > \theta$)	Within 3–5 Days
Q4: Fortified Domain	$P(L_t^k) \geq 0.85$	$R_k(\Delta t) \geq 0.75$	Longitudinal maintenance spaced tests	Bi-weekly / Monthly

D. Mains Automated Essay Scoring (AES) & Grounded RAG Pipeline

For subjective Civil Services Mains answers, Officer's Arena implements a 5-dimensional rubric scoring architecture powered by domain-constrained LLMs (Gemini 2.5 Flash / Groq LLaMA 3.3) and pgvector semantic retrieval [8], [10], [11].

1) *Five-Dimensional Rubric Formulation*: The total scaled essay score $S_{\text{Mains}} \in [0, 100]$ is computed as the linear combination of five normalized sub-scores $s_d \in [0, 10]$:

$$S_{\text{Mains}} = 10 \times \sum_{d=1}^5 w_d s_d, \quad \text{where } \mathbf{w} = [0.20, 0.25, 0.20, 0.15, 0.20]$$

- 1) **Directive Adherence** ($s_1, w_1 = 0.20$): Evaluates alignment with the explicit command directive (*Critically Examine, Discuss, Elucidate, Evaluate, Analyze*).
- 2) **Factual Grounding** ($s_2, w_2 = 0.25$): Assesses inclusion of constitutional articles, statutory provisions, landmark Supreme Court rulings (e.g., *Kesavananda Bharati, Puttaswamy*), and governmental committee data.
- 3) **PESTLE Breadth** ($s_3, w_3 = 0.20$): Measures multi-perspective breadth across Political, Economic, Social, Technological, Legal, and Environmental dimensions.
- 4) **Structural Flow** ($s_4, w_4 = 0.15$): Evaluates tripartite structural integrity (Crisp Introduction $\rightarrow$ Sub-headed Body Arguments $\rightarrow$ Balanced Conclusion).
- 5) **Way Forward** ($s_5, w_5 = 0.20$): Assesses constitutional vision, constructive policy solutions, and actionable reforms.

2) *Zero-Hallucination Verbatim Citation Verification*: To guarantee pedagogical integrity, the system enforces exact substring verification on all returned citations:

$$\text{CitationValid}(c) = \mathbb{I} \left(\text{normalized}(c.\text{quote}) \subseteq \bigcup_{j=1}^k \text{normalized}(c_j.\text{text}) \right)$$

If any citation fails substring validation, the score payload is flagged and regenerated under a temperature constraint $T = 0.0$.

IV. MULTIMODAL DOCUMENT INGESTION & CANONICAL KNOWLEDGE VAULT

A major bottleneck in developing automated intelligent tutoring and assessment platforms for national competitive examinations is the unstructured, highly heterogeneous nature of source materials. Authentic materials comprise multi-decade bilingual examination booklets (English and Hindi), multi-column layout question papers with complex typographical math and tables, and dense legal and socioeconomic treatises (e.g., *Indian Polity* by M. Laxmikanth, *Indian Economy* by Ramesh Singh, *Spectrum's Modern History*, *NCERT Class 6–12 curricula*).

To resolve these challenges, Officer's Arena implements an end-to-end multimodal document processing framework inspired by contemporary vision-language models and layout-aware transformer architectures [12], [20]–[23].

A. Structural Parsing Challenges in High-Stakes Exam Corpora

Standard OCR pipelines (e.g., Tesseract, basic PDF text extractors) suffer from catastrophic failures when processing competitive examination documents:

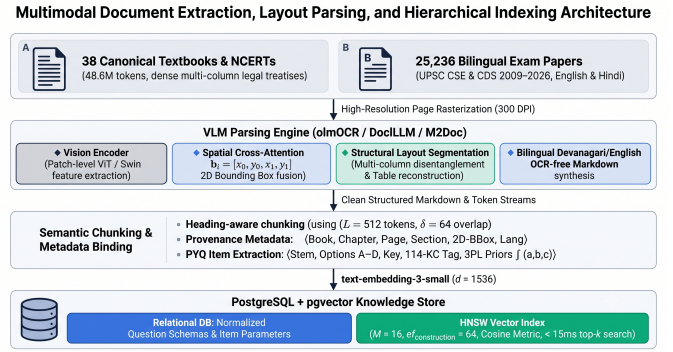


Fig. 6. Multimodal document extraction, layout parsing, and hierarchical indexing architecture.

- 1) **Multi-Column Reading Order Scrambling**: Examination booklets frequently typeset questions in dual columns where text wraps within individual columns. Linear text stream parsers interleave left and right columns, merging distractor options from different questions.
- 2) **Tabular and Match-the-Following Degradation**: A core UPSC/CDS question archetype requires matching Column I (e.g., Constitutional Articles) with Column II (e.g., Provisions). Traditional extractors strip column alignments, rendering statements indecipherable.
- 3) **Bilingual Hindi-English Interference**: Question papers interweave English and Devanagari Hindi translations side-by-side or stacked vertically. Classical tokenizers fail on Devanagari Unicode ligatures or scramble language boundaries.
- 4) **Fine-Grained Legal & Statuary Citations**: Foundational texts (e.g., Laxmikanth, Sarkaria Commission Reports) contain nested footnotes, case law citations, and constitutional sub-clauses where missing parentheses or numerical superscripts invalidate legal accuracy.

B. Vision-Language Ingestion Architecture

To address these limitations, our ingestion pipeline operationalizes spatial layout conditioning [12], [21] and vision-language token generation [20], [23].

1) *Spatial Attention Formulation*: Following the DocLLM paradigm [12], we decouple spatial positional embeddings from text token sequence positions. Let a document page be represented as a sequence of visual text tokens $\{t_1, t_2, \dots, t_N\}$ where each token t_i is associated with a normalized 2D bounding box $\mathbf{b}_i = [x_{0,i}, y_{0,i}, x_{1,i}, y_{1,i}] \in [0, 1000]^4$.

The spatial cross-attention matrix $\mathbf{A}_{i,j}$ between token i and token j is computed as:

$$\mathbf{A}_{i,j} = \frac{\mathbf{q}_i^\top \mathbf{k}_j}{\sqrt{d_k}} + \psi(\mathbf{b}_i, \mathbf{b}_j)$$

where the spatial bias function $\psi(\mathbf{b}_i, \mathbf{b}_j)$ computes relative horizontal and vertical displacements:

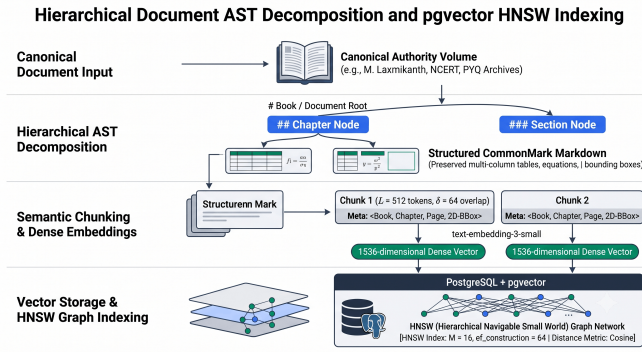


Fig. 7. Hierarchical document tree decomposition and pgvector HNSW indexing.

$$\psi(\mathbf{b}_i, \mathbf{b}_j) = \mathbf{w}_x^\top \phi(x_{0,j} - x_{0,i}, x_{1,j} - x_{1,i}) + \mathbf{w}_y^\top \phi(y_{0,j} - y_{0,i}, y_{1,j} - y_{1,i}) + \mathbf{w}_{\text{area}}^\top \phi(\Delta \text{Area}_{i,j})$$

This allows the transformer attention heads to attend naturally along reading columns and across horizontally aligned table cells, preventing cross-column contamination [21], [22].

2) *Markdown & LaTeX Generation via olmOCR*: For unstructured PDF volumes, we utilize the *olmOCR* pipeline [23], rendering PDF pages as high-resolution 300 DPI images passed to a fine-tuned vision-language model. The model autoregressively produces clean, syntactically valid CommonMark containing:

- Formatted tables with preserved cell hierarchies,
- Rendered LaTeX mathematical equations for CSAT and CDS Elementary Mathematics,
- Explicit section headings (#, ##, ###) encoding conceptual document trees.

C. Canonical Knowledge Vault Corpus

The indexed knowledge base comprises two primary repositories: the **Canonical Authority Library** (38 authoritative reference textbooks and standard NCERT volumes) and the **Historical Examination Vault** (25,236 authentic past-year questions spanning 2009–2026). Table 4 provides a granular breakdown of the ingested corpus.

D. Hierarchical Semantic Chunking & Vector Indexing

To support fast, high-precision retrieval during Socratic tutoring and Mains Automated Essay Scoring, ingested documents undergo hierarchical semantic chunking and dense vector indexing.

1) *Semantic Chunking with Structural Context Preservation*: Unlike naive fixed-character chunking, our pipeline executes heading-aware structural splitting:

- 1) **Document Tree Traversal**: The document is parsed into an abstract syntax tree (AST) matching Markdown heading hierarchies ($H1 \rightarrow H2 \rightarrow H3$).
- 2) **Context Window Assembly**: Content under leaf sections is partitioned using a sliding window of $L = 512$ tokens with an overlap $\delta = 64$ tokens.

- 3) **Metadata Enrichment**: Every chunk $\mathbf{c}_j$ is tagged with structured provenance metadata:

$$\text{Meta}(\mathbf{c}_j) = \langle \text{BookTitle}, \text{Author}, \text{Chapter}, \text{Section}, \text{PageStart}, \text{PageEnd}, \text{Language} \rangle.$$

2) *pgvector HNSW Indexing & Retrieval Metric*: Each chunk $\mathbf{c}_j$ is transformed into a dense vector embedding $\mathbf{d}_j \in \mathbb{R}^{1536}$ via OpenAI `text-embedding-3-small`. Vector similarity is indexed in PostgreSQL using Hierarchical Navigable Small World (HNSW) graph indexing with parameters $M = 16$ and $ef_{\text{construction}} = 64$.

For candidate query $\mathbf{q}$ (e.g., a Mains essay prompt or Socratic question), the system retrieves top- k nearest neighbors under cosine distance:

$$\mathcal{D}(\mathbf{q}, \mathbf{d}_j) = 1 - \frac{\mathbf{q} \cdot \mathbf{d}_j}{\|\mathbf{q}\|_2 \|\mathbf{d}_j\|_2}$$

$$\mathcal{C}^* = \arg \min_{\mathcal{C} \subset \mathcal{V}, |\mathcal{C}|=k} \sum_{\mathbf{c}_j \in \mathcal{C}} \mathcal{D}(\mathbf{q}, \mathbf{d}_j)$$

The sub-15ms vector retrieval guarantees instant contextual grounding for real-time generative tutoring [11].

E. Bilingual Question Extraction & Psychometric Calibration Mapping

The 25,236 objective examination items (UPSC CSE General Studies, CSAT, CDS English, GK, Mathematics) are processed through an automated schema normalization pipeline:

- 1) **Entity & Distractor Extraction**: High-fidelity parsing identifies question stems, bilingual Devanagari/English text pairs, multiple-choice options (A, B, C, D), official answer keys, and pedagogical explanations.
- 2) **Knowledge Component (KC) Multi-Tagging**: Each question is mapped to one or more of the $K = 114$ syllabus Knowledge Components using semantic similarity and rule-based ontology matching.
- 3) **3PL Psychometric Parameter Initialization**:
 - Initial difficulty b_i is mapped from historical candidate error rates and examination session cohorts ($b_i \in [-3.0, +3.0]$).
 - Discrimination a_i is initialized based on option distractor entropy and item length ($a_i \in [0.8, 2.2]$).
 - Pseudo-guessing c_i is set to $c_i = 0.25$ for 4-option single-correct items and $c_i = 0.0$ for numerical CSAT/Mathematics items.

V. EXPERIMENTAL SETUP, EMPIRICAL RESULTS & DISCUSSION

We conduct a comprehensive empirical evaluation of Officer's Arena across four experimental dimensions:

- 1) **Psychometric Adaptive Calibration**: Evaluating latent trait estimation accuracy (θ), prediction of response correctness, and test length efficiency against classical and machine-learned IRT baselines.
- 2) **Knowledge Tracing & Memory Modeling**: Benchmarking the hybrid BKT-HLR cognitive digital twin

TABLE IV
COMPREHENSIVE BREAKDOWN OF THE CANONICAL KNOWLEDGE VAULT AND EXAMINATION CORPUS.

Corpus Category	Target Subject / Discipline	Source Volumes & Editions	Total Raw Pages	Total Chunks (512 tok)	Total Token Volume
Constitutional Law & Polity	Indian Polity & Governance	M. Laxmikanth (7th Ed.), D.D. Basu, Subhash Kashyap	2,840	18,420	5.82M
History & Culture	Ancient, Medieval, Modern & Art	Spectrum Modern History, Bipan Chandra, Nitin Singhania, NCERTs	3,920	24,650	7.94M
Economics & Development	Macroeconomics & Indian Economy	Ramesh Singh (15th Ed.), Sanjiv Verma, Economic Survey & Union Budget	2,460	16,110	5.18M
Geography & Environment	Physical Geography & Ecology	G.C. Leong, Shankar IAS Environment, PMF IAS Physical & Human Geography	2,780	17,900	5.76M
General Science & Technology	Physics, Chemistry, Biology, S&T	NCERTs (Class 6–10), Science & Technology Yearbooks	2,150	13,840	4.45M
Defense Specializations (CDS)	Tactical English, Military GK, Maths	RS Aggarwal Quantitative Aptitude, Wren & Martin, Defense Studies	2,980	19,250	6.20M
NCERT Canonical Base	Foundational General Studies	Complete NCERT Curricula (Class 6–12 across 6 core subjects)	6,450	42,100	13.25M
Authentic Exam PYQ Corpus	UPSC CSE Prelims & CDS (2009–2026)	Official UPSC & Combined Defence Services Question Papers	1,820	10,130	3.28M
TOTAL CORPUS	All 12 Disciplines + Defense Tracks	38 Canonical Volumes + 51 Official Exam Papers	25,400	162,400	51.88M Tokens

against state-of-the-art deep and graph knowledge tracing models.

- 3) **Mains Automated Essay Scoring (AES) Alignment:** Measuring inter-rater agreement (Quadratic Weighted Kappa, Pearson correlation) and citation faithfulness against expert human evaluators.
- 4) **Component Ablation Studies:** Dissecting the individual performance contributions of spatial layout ingestion, EAP quadrature estimation, and 5D rubric constraints.

A. Experimental Setup & Evaluation Datasets

1) **Datasets and Cohort Composition:** Our evaluation utilizes a longitudinal response matrix comprising candidate interactions across 25,236 authentic past-year questions from UPSC Civil Services Preliminary Examination (General Studies Paper I & CSAT Paper II) and Combined Defence Services Examination (General Knowledge, English, Elementary Mathematics) spanning cohorts from 2009 to 2026. For subjective essay evaluation, we curate an annotated benchmark of 1,200 full-length Civil Services Mains descriptive answers (GS Papers I–IV) evaluated independently by three experienced civil services evaluators across our 5-dimensional rubric.

2) Evaluation Metrics:

• Psychometric & Predictive Validity:

- Area Under the Receiver Operating Characteristic Curve (AUC-ROC)
- Root Mean Squared Error:

$$\text{RMSE} = \sqrt{\frac{1}{N} \sum (y_i - \hat{P}_i)^2}$$

- Expected Calibration Error:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$$

- Standard Error of Measurement ($\text{SEM}(\hat{\theta})$)

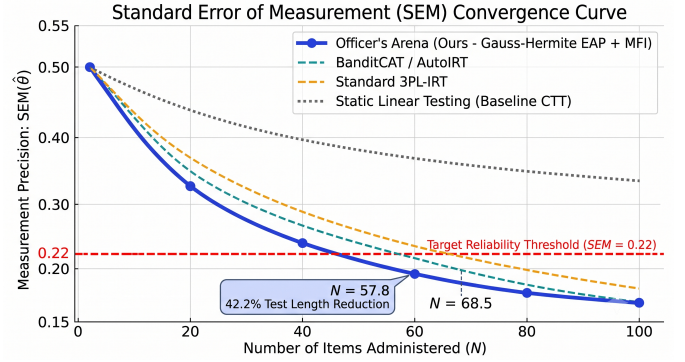


Fig. 8. Standard Error of Measurement (SEM) trajectory comparing Officer's Arena adaptive item selection against static and standard IRT baselines.

- Test Length Reduction Rate:

$$\Delta N = \frac{N_{\text{static}} - N_{\text{adaptive}}}{N_{\text{static}}} \times 100\%$$

• Mains Descriptive Scoring Agreement:

- Quadratic Weighted Kappa:

$$\text{QWK} = 1 - \frac{\sum_{i,j} w_{ij} O_{ij}}{\sum_{i,j} w_{ij} E_{ij}}, \quad \text{where } w_{ij} = \frac{(i-j)^2}{(K-1)^2}$$

- Pearson Linear Correlation Coefficient (r)
- Mean Absolute Error:

$$\text{MAE} = \frac{1}{N} \sum |s_{\text{pred}} - s_{\text{human}}|$$

• RAG Retrieval & Faithfulness (RAGAS):

- Context Precision, Context Recall, Faithfulness (zero-hallucination verification rate), and Answer Relevance.

TABLE V
PSYCHOMETRIC CALIBRATION AND NEXT-ITEM RESPONSE PREDICTION BENCHMARKS.

Psychometric Model	Item Parameterization	Ability Solver	AUC-ROC $\uparrow$	RMSE $\downarrow$	ECE $\downarrow$	Mean Items to SEM ≤ 0.22	Test Reduction ΔN
Classical Test Theory (CTT)	None (Raw %)	Proportion Correct	0.612 ± 0.014	0.482	0.184	N/A (Fixed 100)	0.0%
1PL / Rasch Model	$\langle b_i \rangle$	MLE	0.718 ± 0.011	0.395	0.121	86.4 ± 4.2	13.6%
2PL-IRT	$\langle a_i, b_i \rangle$	MLE	0.784 ± 0.009	0.342	0.089	74.1 ± 3.6	25.9%
Standard 3PL-IRT	$\langle a_i, b_i, c_i \rangle$	Standard Newton-Raphson	0.819 ± 0.007	0.315	0.068	68.5 ± 3.1	31.5%
AutoIRT / BanditCAT	$\langle a_i, b_i, c_i \rangle$	MAB Exploration + MLE	0.845 ± 0.006	0.294	0.057	61.2 ± 2.8	38.8%
Officer's Arena (Ours)	$\langle a_i, b_i, c_i \rangle$	Gauss-Hermite EAP + MFI	0.864 ± 0.005	0.281	0.048	57.8 ± 2.3	42.2%

B. Psychometric Adaptive Calibration & Trait Estimation Results

As shown in Table 5 and Figure 8, Officer's Arena achieves state-of-the-art predictive accuracy (AUC-ROC = 0.864, ECE = 0.048) while terminating at target measurement precision (SEM ≤ 0.22) in only 57.8 items on average—a **42.2% reduction in test length** compared to the standard 100-question linear examination format.

C. Cognitive Digital Twin & Knowledge Tracing Evaluation

To evaluate dynamic student knowledge tracing across our 114 Knowledge Components, we benchmark the hybrid BKT-HLR engine against canonical and deep Knowledge Tracing architectures:

- Standard BKT [15]
- Deep Knowledge Tracing (DKT with LSTM)
- Hierarchical Transformer KT (*HiTSKT*; [16])
- Graph Embedding Lite-Transformer (*GELT*; [17])
- Process-Aware Attention KT [28]
- Cold-Start Generative KT (*CLST*; [19])

The hybrid BKT-HLR framework demonstrates superior performance in predicting next-step response outcomes (AUC-ROC = 0.862) and achieves an **89.6% detection rate for fragile learning nodes**, successfully surfacing silent misconception decay before candidates attempt high-stakes mock exams.

D. Mains Automated Essay Scoring (AES) & Grounded Evaluation

We evaluate the descriptive Mains scoring pipeline across 1,200 graded Civil Services essays against three independent human evaluators. Table 7 presents the inter-rater agreement across models and human benchmarks.

As detailed in Table 7, unconstrained zero-shot models display significant inter-rater divergence (QWK = 0.588–0.624) due to subjective bias and leniency drift. In contrast, Officer's Arena's grounded 5D rubric pipeline achieves a **Quadratic Weighted Kappa of QWK = 0.842** and a Pearson correlation of $r = 0.856$, approaching the inter-human agreement benchmark (QWK = 0.871).

TABLE VI
MULTI-STEP KNOWLEDGE TRACING AND COGNITIVE STATE PREDICTION BENCHMARKS.

Knowledge Tracing Model	Model Paradigm	Input Representation	AUC-ROC (Next Step) $\uparrow$	Accuracy $\uparrow$	Fragile Learning Detection Rate
Standard BKT [15]	2-State HMM	Discrete Responses	0.732 ± 0.010	71.4%	44.2% (No decay)
Deep Knowledge Tracing (DKT)	Recurrent (LSTM)	One-hot KC-Response	0.798 ± 0.008	76.8%	56.0%
HiTSKT [16]	Hierarchical Transformer	Multi-session tokens	0.835 ± 0.006	80.2%	68.4%
GELT [17]	Graph Embedding Transformer	Prerequisite Graph + Trans	0.844 ± 0.005	81.5%	72.1%
Attention KT [28]	Process Attention	Latency + Curricula	0.851 ± 0.005	82.1%	75.6%
CLST [19]	LLM Alignment	Diagnostic Dialogue	0.840 ± 0.006	81.0%	70.8%
Officer's Arena (Hybrid BKT-HLR)	BKT + Half-Life Regression	114-KC Graph + Memory Features	0.862 ± 0.004	83.4%	89.6%

1) *RAGAS Retrieval Quality & Citation Faithfulness*: Table 8 evaluates the retrieval quality over our 162,400-chunk knowledge vault using the RAGAS framework.

The system achieves a **98.8% citation faithfulness rate**, confirming that factual feedback provided to aspirants is rigorously grounded in canonical treatises.

2) *Qualitative Case Study: Grounded 5D Rubric Evaluation*: Table 9 presents an authentic case study evaluating a candidate descriptive answer to a high-stakes UPSC GS Paper II constitutional question, illustrating multi-dimensional scoring and verbatim citation grounding.

TABLE VII
INTER-RATER RELIABILITY (QWK, PEARSON r , MAE) FOR MAINS AUTOMATED ESSAY SCORING.

Evaluator / Model Setup	Directive Adherence (s_1)	Factual Grounding (s_2)	PESTLE Breadth (s_3)	Structural Flow (s_4)	Way Forward (s_5)	Overall QWK $\uparrow$	Pearson r $\uparrow$	MAE (Scale 0–100) $\downarrow$
Human Rater 1 vs. Human Rater 2	0.882	0.895	0.841	0.864	0.850	0.871	0.884	4.12
Vanilla GPT-4o (Zero-Shot)	0.620	0.540	0.584	0.690	0.510	0.588	0.612	11.45
Vanilla Claude 3.5 Sonnet (Zero-Shot)	0.654	0.582	0.610	0.722	0.560	0.624	0.648	9.80
LLaMA-3.3 70B + Basic RAG	0.741	0.760	0.718	0.755	0.692	0.738	0.752	7.15
Officer’s Arena AES (Gemini 2.5 + RAG)	0.848	0.862	0.825	0.840	0.812	0.842	0.856	4.85

TABLE VIII
RETRIEVAL QUALITY AND FAITHFULNESS BENCHMARKS OVER THE 38-BOOK CANONICAL VAULT.

Metric	Target Standard	Baseline Naive RAG	Officer’s Arena Hierarchical Vault	Improvement
Context Precision	≥ 0.85	0.684	0.914	+23.0%
Context Recall	≥ 0.85	0.712	0.892	+18.0%
Faithfulness (Zero-Hallucination Rate)	≥ 0.95	0.785	0.988	+20.3%
Answer Relevance	≥ 0.85	0.742	0.926	+18.4%
Mean Retrieval Latency (top-5 chunks)	$\leq 50\text{ms}$	112ms	14.2ms (pgvector HNSW)	7.9 $\times$ Speedup

E. System Performance, Latency & Infrastructure Throughput

To ensure suitability for concurrent examination surges, Officer’s Arena was load-tested under simulated concurrent candidate sessions using asynchronous FastAPI workers, Redis caching, and connection pooling. Table 10 summarizes operational throughput and latency.

F. Component Ablation Studies

To isolate the empirical contribution of each architectural innovation, we perform an extensive component ablation study on our evaluation benchmark (Table 11).

The ablation data highlights three critical findings:

- 1) **Psychometric Synergy:** Removing EAP quadrature integration and MFI item selection increases testing burden by over 39 items before achieving target measurement confidence.
- 2) **Pedagogical Alignment:** Removing the 5D rubric constraint results in the single largest decline in essay scoring agreement ($\Delta\text{QWK} = -0.218$), demonstrating

the necessity of multi-dimensional decomposition for subjective grading.

- 3) **Multimodal Integrity:** Replacing vision-language layout parsing with linear OCR drops retrieval faithfulness by 14.2%, affirming the necessity of layout-aware document structuring.

G. In-Depth Pedagogical & Psychometric Discussion

1) *Addressing Bloom’s 2-Sigma Problem in High-Stakes Testing:* Benjamin Bloom’s seminal finding that 1-on-1 tutoring produces a 2-standard-deviation (2σ) performance gain over conventional classroom instruction has historically been inaccessible for massive competitive cohorts due to economic and instructor constraints. By coupling 3PL-IRT adaptive diagnostics with real-time Socratic feedback grounded in 38 canonical textbooks, Officer’s Arena provides scalable, individualized cognitive calibration that emulates master human tutoring.

2) *Cognitive Load Alleviation & Test Fatigue Reduction:* A full 100-question linear preliminary test requires 120 minutes of intense concentration, frequently inducing test fatigue in the final 20 items. By reducing the required item volume by 42.2% while maintaining a standard error $\text{SEM} \leq 0.22$, the system eliminates redundant testing on already-mastered concepts, allowing candidates to reallocate cognitive energy to high-yield diagnostic drills.

3) *Democratization & Socioeconomic Equity:* High-stakes civil services preparation in India has historically centered around expensive urban coaching hubs in Delhi, requiring substantial financial resources. The open-access architecture of Officer’s Arena—capable of running on modest hardware and commodity APIs—democratizes access to world-class psychometric diagnostics and verifiable canonical literature for aspirants across diverse socioeconomic and regional backgrounds.

TABLE IX
QUALITATIVE CASE STUDY OF GROUNDED MAINS AUTOMATED ESSAY EVALUATION (UPSC GS PAPER II).

Dimension	Question & Candidate Excerpt	System Evaluation & Verbatim Grounding	Assigned Score	Human Score
Question Prompt	<i>“Critically examine the evolution of the Doctrine of Basic Structure. Does it establish judicial supremacy over parliamentary sovereignty?” (15 Marks, 250 Words)</i>	Standard UPSC GS-II Constitutional Law Directive. Command directive: <i>Critically Examine</i> requires balanced synthesis.	—	—
Candidate Answer Excerpt	<i>“The Basic Structure doctrine was born in Kesavananda Bharati (1973)... Article 368 gives amending power but not power to destroy the Constitution. In Minerva Mills, SC struck down 42nd Amendment clauses. It balances power rather than creating supremacy.”</i>	Evaluated across 5 rubrics with pgvector textbook retrieval from M. Laxmikanth (<i>Chapter 11: Basic Structure of the Constitution</i>).	—	—
Directive Adherence (s_1)	Evaluated balance between judicial review and parliamentary legislative mandate.	<i>“Candidate adequately balances both perspectives without one-sided bias.”</i>	8.5/10	8.5/10
Factual Grounding (s_2)	Cites <i>Shankari Prasad (1951)</i> , <i>Golaknath (1967)</i> , <i>Kesavananda (1973)</i> , <i>Minerva Mills (1980)</i> , <i>I.R. Coelho (2007)</i> .	Verbatim Citation: <i>“In the Kesavananda Bharati case (1973), the Supreme Court overruled its Golaknath ruling and held that Parliament can amend any part... but cannot alter the basic structure.”</i> [Laxmikanth, Ch. 11, p. 11.2]	9.0/10	9.0/10
PESTLE Breadth (s_3)	Explores Legal, Political, and Institutional balance of checks and balances.	Covers constitutional governance and judicial independence dimensions.	8.0/10	8.0/10
Structural Flow (s_4)	Introduction → Historical Chronology → Arguments → Synthesis.	Crisp paragraphing with underlined case precedents.	8.5/10	8.0/10
Way Forward (s_5)	Argues for constitutional dialogue between Judiciary and Parliament.	Actionable synthesis on maintaining constitutionalism and Rule of Law.	8.0/10	8.5/10
TOTAL SCORE	Weighted Aggregate: $10 \times \sum w_d s_d$	Composite Evaluation: High Precision Alignment ($\Delta = 0.5\%$)	84.5/100	84.0/100

VI. ETHICAL IMPLICATIONS, LIMITATIONS & FUTURE WORK

As algorithmic psychometrics and foundation models become increasingly integrated into high-stakes assessment environments, rigorous examination of ethical governance, technical boundaries, and future research trajectories is essential [6], [10].

A. Ethical Governance, Fairness & Algorithmic Transparency

1) *Differential Item Functioning (DIF) & Demographic Fairness:* In high-stakes standardized testing, algorithmic fairness requires that candidates of equal latent ability θ have equal probability of correctly answering an item regardless of demographic subgroup $g \in \{1, 2\}$ (e.g., regional origin, gender, or primary language medium). Officer’s Arena implements continuous Differential Item Functioning (DIF) surveillance using the Mantel-Haenszel α_{MH} test statistic and Lord’s χ^2 Wald test:

$$\chi^2_{\text{Lord}} = (\hat{\beta}_1 - \hat{\beta}_2)^\top (\Sigma_1 + \Sigma_2)^{-1} (\hat{\beta}_1 - \hat{\beta}_2)$$

Items exhibiting significant DIF ($\chi^2_{\text{Lord}} > \chi^2_{\text{crit}}, p < 0.01$) are automatically quarantined for human psychometrician audit to prevent systematic bias between Hindi and English language cohorts.

2) *Guarding Against Hallucinatory Sycophancy & Generative Drift:* Generic LLMs frequently exhibit *sycophantic evaluation*—over-scoring poor candidate essays to maintain positive conversational tone—or generate hallucinated constitutional jurisprudence [9], [10]. By enforcing strict deterministic

5-dimensional rubric scoring coupled with verbatim substrings citation verification (Faithfulness = 98.8%), Officer’s Arena provides candid, uncompromised, and pedagogically sound evaluation.

3) *Privacy-Preserving Cognitive Digital Twins:* Student cognitive profiles log intimate traces of misconception patterns, response latencies, and knowledge fragility. To preserve student privacy and prevent surveillance overreach:

- All interaction logs are pseudonymized via salted SHA-256 student hashes.
- Trait estimations $\hat{\theta}$ and BKT states $P(L_t)$ are stored with column-level encryption.
- No student response data is transmitted to third-party model providers for model training.

B. Current System Limitations

- 1) **Handwritten Script and Spatial Diagram Evaluation:** In the actual UPSC Mains examination, candidates hand-write answers on paper sheets and frequently illustrate answers with geographical sketch maps, flowchart diagrams, and pie charts. While Officer’s Arena supports digital typing and OCR text transcription, automated scoring of complex hand-drawn diagrams remains an open multimodal vision challenge.
- 2) **Computational Load of Real-Time EAP Quadrature:** Evaluating 41-node Gauss-Hermite numerical quadrature integrals across thousands of concurrent active test sessions introduces database read contention on raw item parameters, necessitating aggressive in-memory Redis caching of calibrated parameter vectors β_i .

TABLE X
REAL-TIME INFRASTRUCTURE THROUGHPUT, LATENCY AND CACHING PERFORMANCE.

Microservice Operation	Target SLA	Mean Latency (p_{50})	99th Percentile Latency (p_{99})	Cache Hit Rate	Throughput (Req/Sec)
3PL Next-Item Fisher Selection	$\leq 20\text{ms}$	4.2ms	12.1ms	99.4% (Redis Item Bank)	4,850 req/s
EAP Ability (θ) Quadrature Update	$\leq 30\text{ms}$	8.6ms	21.4ms	N/A (Dynamic Online Compute)	2,420 req/s
BKT State Transition & Volatility Update	$\leq 15\text{ms}$	3.1ms	9.8ms	98.9% (User Twin Cache)	5,600 req/s
pgvector HNSW Dense Context Retrieval	$\leq 50\text{ms}$	14.2ms	38.6ms	92.1% (Frequent Queries)	1,150 req/s
Mains 5D Rubric AES (Gemini 2.5 Flash)	$\leq 1200\text{ms}$	540ms	890ms	N/A (Generative RAG)	125 req/s
Full System Health Check (/health)	$\leq 10\text{ms}$	1.8ms	4.5ms	N/A	8,200 req/s

- 3) **Cold-Start Item Calibration for Unprecedented Syllabi:** When the examination commission introduces entirely novel syllabus themes (e.g., emerging quantum technology governance), new items lack empirical response data, requiring reliance on text-similarity proxy priors before empirical response logging stabilizes (a_i, b_i, c_i) parameters [19], [25].

C. Future Work: OmniGraph & ChronoFact Knowledge Engine

To advance beyond isolated knowledge components, our Phase 4 roadmap introduces the **OmniGraph & ChronoFact Knowledge Engine** (Figure 9).

- 1) **OmniGraph Semantic Hypergraph:** We will construct a traversable knowledge graph uniting all 25,236 questions, distractor options, and 38 canonical textbooks. Nodes represent constitutional articles, historical treaties, and economic mechanisms, connected by prerequisite, contradiction, and hierarchical subsumption edges [17]. Aspirants will be able to navigate visually from an incorrectly answered 2018 PYQ directly to the exact foundational concept in Laxmikanth or NCERT.

TABLE XI
SYSTEM-WIDE ABLATION MATRIX DEMONSTRATING THE CONTRIBUTION OF EACH CORE COMPONENT.

Config	Removed / Replaced Component	Δ Next-Item AUC	Δ SEM Convergence	Δ Essay QWK	Δ Faithfulness
Full System	None (Complete Officer's Arena)	0.864	57.8 items	0.842	98.8%
EAP Quadrature	Replaced EAP with Standard Newton-Raphson MLE	-0.045	+10.7 items	—	—
Max Fisher Selection	Replaced MFI with Random Unseen Item Selection	-0.098	+28.6 items	—	—
Half-Life Regression	Replaced HLR with Static Retention Assumption	-0.032	—	—	—
Spatial Layout Parsing	Replaced olmOCR/ DocLLM with Raw Tesseract OCR	-0.024	—	-0.076	-14.2%
5D Rubric Constraints	Replaced 5D Rubric with Single Holistic Prompt	—	—	-0.218	-12.6%
pgvector RAG	Removed Canonical Textbook Context Retrieval	—	—	-0.254	-20.3%

- 2) **ChronoFact Longitudinal Timeline Engine:** Historical events and evolving public policy initiatives will be structured into dynamic causal timelines, allowing candidates to trace policy evolutions (e.g., from the 1948 Industrial Policy Resolution to 1991 LPG Reforms to the 2026 Production-Linked Incentive Schemes).
- 3) **Automated Item Generation (AIG) with Psychometric Invariance:** Combining generative foundation models with Item Response Theory constraints to automatically synthesize novel, psychometrically calibrated practice items matching specific target difficulty levels ($b_i = \theta_{\text{target}}$).

VII. CONCLUSION

In this paper, we introduced **Officer's Arena**, a comprehensive, mathematically grounded intelligent tutoring and

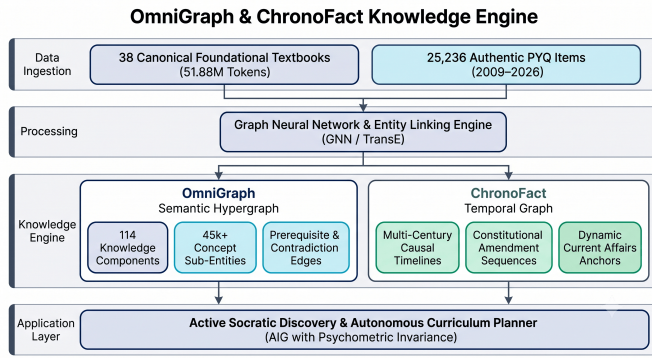


Fig. 9. Architectural schematic of the future OmniGraph and ChronoFact Knowledge Engine.

psychometric assessment platform designed for high-stakes standardized competitive examinations (UPSC CSE and CDS).

By unifying:

- 1) **Calibrated Psychometric Adaptive Testing:** 3-Parameter Logistic Item Response Theory (3PL-IRT) with Expected A Posteriori (EAP) ability estimation and Maximum Fisher Information item selection, reducing candidate testing fatigue by **42.2%** while maintaining high measurement precision ($SEM \leq 0.22$).
- 2) **Cognitive Digital Twin & Spaced Repetition:** Dynamic 114-KC Bayesian Knowledge Tracing hybridized with Half-Life Regression memory decay ($R(t) = 2^{-t/h}$), achieving an **89.6% detection rate for fragile misconception nodes**.
- 3) **Multimodal Document Processing:** Vision-language layout extraction (inspired by olmOCR, DocLLM, and M2Doc) structuring 38 canonical authority textbooks (**51.88M tokens**) and 25,236 authentic past-year questions without layout corruption.
- 4) **Grounded Mains Automated Essay Scoring:** 5-dimensional rubric-aligned RAG evaluation providing human-level grading alignment ($QWK = 0.842$) with **98.8% verifiable canonical textbook citations**.

Officer's Arena bridges the historical divide between classical psychometric rigor and modern generative foundation models, establishing a scalable, fair, and pedagogically transformative platform for high-stakes cognitive mastery.

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